Minimum Feedback Arc Sets in Rotator Graphs

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Abstract

A feedback vertex set/arc set (abbreviated to FVS/FAS) is a vertex/arc subset of a graph whose removal induces the remaining graph acyclic. A minimum FVS/FAS is an FVS/FAS which contains the smallest number of vertices/arcs. Hsu and Lin [8] first proposed an algorithm which can find a minimum FVS in rotator graphs. In this paper, we present two efficient algorithms both of which construct FAS's for a rotator graph with n! nodes in O(nn!) time and also prove that the FAS's are minimum. In other words, these algorithms derive the optimal results with linear time complexity in terms of the number of arcs in the rotator graph.

Keywords: *Feedback arc set; Rotator graph; Generator; Directed cycle*

1 Introduction

A feedback vertex set/arc set (abbreviated to FVS/FAS) is a vertex/arc subset of a graph whose removal induces the remaining graph acyclic. The FVS/FAS researches can be applied to deadlock prevention in operating systems [13] and scheduling [4]. A minimum FVS/FAS is an FVS/FAS which contains the smallest number of vertices/arcs. The both problems of finding a minimum FVS for general graphs and FAS for general directed graphs are NP-hard problems [6]. Therefore, many researches about FVS have been proposed for special graphs such as mesh and butterfly [1,11], hypercube [5], star graph [15], rotator graph [8], shuffle-based interconnection networks [9], and so on. FVS can be studied for both undirected and directed graphs, but only few FAS researches have been proposed for directed graphs such as reducible flow graphs[12], cyclically reducible flow graphs[14], tournaments[2] and bipartite tournaments[7].

Rotator graph, a directed graph and a family of Cayley graph, is first proposed in 1992 [3]. In recent years, many related papers have been published because of the rich topological properties of the rotator graph, such as symmetric structure, recursive construction, low diameter, unique shortest path routing, and so on. In 2006, Hsu and Lin [8] first proposed an algorithm for finding a minimum FVS in rotator graphs. Subsequently, Kuo and Hsu [10] proposed an algorithm for finding a minimum FVS in rotator graphs with linear time complexity in terms of the number of nodes in the rotator graph. In this paper, we present two efficient algorithms which find minimum FAS's in rotator graphs also with linear time complexity in terms of the number of arcs in the rotator graph.

2 Preliminaries

Let $N = \{1, 2,\ldots, n\}$ and $s_1s_2s_3\ldots s_n$ be a permutation, where $s_i \in N$ for $1 \le i \le n$. A rotator graph of scale n, denoted by n-rotator graph, has $n!$ nodes labeled with a unique permutation $s_1s_2s_3\ldots s_n$. Let $v = s_1s_2s_3\ldots s_n$ be a node in an n-rotator graph and s_i be the i^{th} symbol of v. Generator g_i inserts the first symbol of a node to the i^{th} position to form an adjacent node of v. Node $v * g_i$ denotes the adjacent node of applying g_i to node v. Node v links to adjacent nodes $v * g_i$ by the outgoing edges represented as generators g_i for $2 \le i \le n$. A 3-rotator graph can be represented by either Fig. 1 or Fig. 2 where edge g_2 is represented by directed and undirected edges, respectively. Note that node and vertex are identical in this paper, so are arc and edge.

Definition 1. (v, g_i) denotes the outgoing arc g_i of node v.

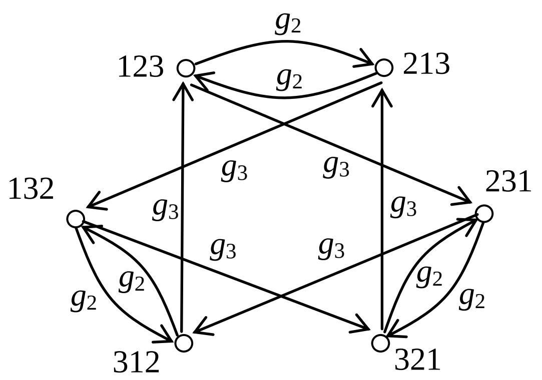

Fig. 1. A 3-rotator graph with directed edge g_2.

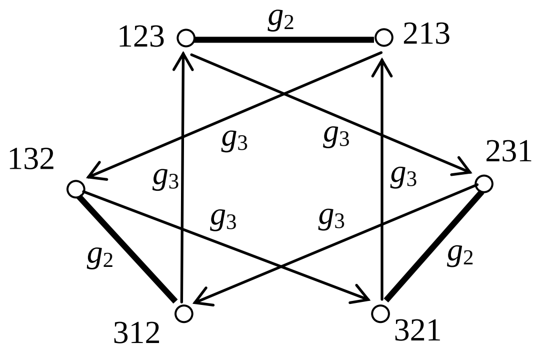

Fig. 2. A 3-rotator graph with undirected edge g_2.

Definition 2. Node $v * \delta$ denotes the resultant of applying δ to node v, where δ can be a single generator or a sequence of generators.

Definition 3. Let g_i and g_j be two generators. If $i > j$, we call that g_i is a higher dimension generator than g_j and denoted by $g_i \succ g_j$.

Definition 4. Let $V_{i,j}$ of a rotator graph be the set of all nodes whose i^{th} symbol is j.

For instance, $V_{2,1}$ = {2134, 2143, 3124, 3142, 4123, 4132} for a 4-rotator graph.

Definition 5. A directed cycle C in a rotator graph can be represented as $\pi(v, R)$, where v is a node of C and R is the generator sequence of C starting from v. Two directed cycles C_1 and C_2 are said to be equivalent, denoted by $C_1 = C_2$, if C_1 and C_2 contain the same nodes and the same arcs.

For instance, $\pi(1234, g_4\, g_3\, g_4\, g_3\, g_4\, g_3\, g_4\, g_3)$ is a directed cycle of 1234 $_{(4)}\!\rightarrow$ 2341 $_{(3)}\!\rightarrow$ 3421 $_{(4)}\!\rightarrow$ 4213 $_{(3)}\!\rightarrow$ 2143 $_{(4)}\!\rightarrow$ 1432 $_{(3)}\!\rightarrow$ 4312 $_{(4)}\!\rightarrow$ 3124 $_{(3)}\!\rightarrow$ 1234. Clearly, a directed cycle can be represented in different expressions by starting from different nodes.

Definition 6. Let δ_1 and δ_2 be two sequences of generators. A generator sequence $R = \delta_1 \oplus \delta_2$ denotes the concatenation of two sub sequences δ_1 and δ_2. A generator sequence $R \otimes \delta_1 = \delta_2$, denotes removing the leading sub sequence δ_1 from R.

The following lemma 1 is cited from [10] for the completeness of the whole proof.

Lemma 1. If directed cycle $C_1 = \pi(v, R)$ where $R = \delta_1 \oplus \delta_2$ and directed cycle $C_2 = \pi(v * \delta_1, \delta_2 \oplus \delta_1)$, then $C_1 = C_2$.

Proof. $C_2 = \pi(v * \delta_1, \delta_2 \oplus \delta_1)$

$= \pi(v * \delta_1 * \delta_2, (\delta_2 \oplus \delta_1) \otimes \delta_2 \oplus \delta_2)$

$= \pi(v, \delta_1 \oplus \delta_2)$

$= \pi(v, R)$

$= C_1$ □

For example, let v =1234 and $R = (g_2\, g_4\, g_3) \oplus (g_3\, g_4\, g_2\, g_3)$. Suppose directed cycle $C_1 = \pi(v, R)$ = 1234 $_{(2)}\!\rightarrow$ 2134 $_{(4)}\!\rightarrow$ 1342 $_{(3)}\!\rightarrow$ 3412 $_{(3)}\!\rightarrow$ 4132 $_{(4)}\!\rightarrow$ 1324 $_{(2)}\!\rightarrow$ 3124 $_{(3)}\!\rightarrow$ 1234. Directed cycle $C_2 = \pi(v * (g_2\, g_4\, g_3), (g_3\, g_4\, g_2\, g_3) \oplus (g_2\, g_4\, g_3))$ = $\pi(3412, g_3\, g_4\, g_2\, g_3\, g_2\, g_4\, g_3)$ = 3412 $_{(3)}\!\rightarrow$ 4132 $_{(4)}\!\rightarrow$ 1324 $_{(2)}\!\rightarrow$ 3124 $_{(3)}\!\rightarrow$ 1234 $_{(2)}\!\rightarrow$ 2134 $_{(4)}\!\rightarrow$ 1342 $_{(3)}\!\rightarrow$ 3412. Clearly, $C_1 = C_2$ because they contain the same nodes and arcs.

Lemma 2. Let $C = \pi(v, R)$ be a directed cycle in an n-rotator graph, where $v = s_1s_2s_3\ldots s_n$ and g_j is the first generator in R. For all symbol s_i with $1 \le i \le j$, there exists at least one node in the directed cycle C whose first symbol is s_i.

Proof. The directed cycle C starts from and ends with the node v, and the next node of v in C is $v * g_j$, which is $s_2s_3\ldots s_j\, s_1 s_{j+1}\ldots s_n$, so all symbols of s_2, s_3, …, and s_j must be shifted to the first position in at least one of the nodes in the directed cycle C such that the symbol s_i, for $2 \le i \le j$, can be relocated to the i^{th} symbol position of v. □

For instance, let $C = \pi(v, R)$, where v = 2341 and $R = g_3\, g_4\, g_3\, g_4\, g_3\, g_4\, g_3\, g_4$. Since the first generator is g_3, symbols s_1 = 2, s_2 = 3and s_3 = 4 exist in the first symbol of the nodes in the directed cycle C = **2**341 $_{(3)}\!\rightarrow$ **3**421 $_{(4)}\!\rightarrow$ **4**213 $_{(3)}\!\rightarrow$ **2**143 $_{(4)}\!\rightarrow$ 1432 $_{(3)}\!\rightarrow$ **4**312 $_{(4)}\!\rightarrow$ **3**124 $_{(3)}\!\rightarrow$ 1234 $_{(4)}\!\rightarrow$ **2**341.

Lemma 3. Let $C = \pi(v, R)$ be a directed cycle in an n-rotator graph, where $v = s_1s_2s_3\ldots s_n$ and g_k is the highest dimension generator in R. For any node v' in the directed cycle C, the i^{th} symbol of $v' \in \{s_1, s_2,\ldots,s_k\}$ for $1 \le i \le k$.

Proof. Case $k = n$. Obviously, all symbols of $v' \in \{s_1, s_2, \ldots, s_n\}$, and it is true. Case $k < n$. Because g_k is the highest dimension generator in R and $(v' * g_k)$ denotes that generator g_k inserts the first symbol of node v' into the k^{th} position, the j^{th} symbol of all nodes in the directed cycle C is s_j for $(k + 1) \le j \le n$. In other words, the i^{th} symbol of all nodes in the directed cycle $C \in \{s_1, s_2, \ldots, s_k\}$ for $1 \le i \le k$. □

Lemma 4. Let $C = \pi(v, R)$ be a directed cycle in an n-rotator graph, where $v = s_1s_2s_3\ldots s_n$ and g_k is the highest dimension generator in R. For all symbol s_i with $1 \le i \le k$, there exists at least one node in the directed cycle C whose first symbol is s_i.

Proof. By Lemma 1, there exists a directed cycle $C_1 = \pi(v_1, R_1) = C$, where $v_1 = s'_1s'_2s'_3\ldots s'_n$, and the first generator of R_1 is the highest dimension g_k. By Lemma 2, for all symbol s'_i with $1 \le i \le k$, there exists at least one node in directed cycle C_1 whose first symbol is s'_i. By Lemma 3, all symbol $s'_i \in \{s_1, s_2, \ldots, s_k\}$ for $1 \le i \le k$, so for all symbol s_i with $1 \le i \le k$, there exists at least one node in directed cycle C_1 whose first symbol is s_i □

For example, let $C = \pi(v, R)$ be a directed cycle in an n-rotator graph, where $v = 23415$ and $R = g_3\, g_4\, g_3\, g_4\, g_3\, g_4\, g_3\, g_4$. Since the highest generator is g_4, symbols $s_1 = 2$, $s_2 = 3$, $s_3 = 4$, and $s_4 = 1$ exist in the first symbol of nodes in the directed cycle C = $\underline{\mathbf{2}}3415 \xrightarrow{(3)} \underline{\mathbf{3}}4215 \xrightarrow{(4)} \underline{\mathbf{4}}2135 \xrightarrow{(3)} \underline{\mathbf{2}}1435 \xrightarrow{(4)} \underline{\mathbf{1}}4325 \xrightarrow{(3)} \underline{\mathbf{4}}3125 \xrightarrow{(4)} \underline{\mathbf{3}}1245 \xrightarrow{(3)} \underline{\mathbf{1}}2345 \xrightarrow{(4)} \underline{\mathbf{2}}3415$.

Definition 7. Let $C = \pi(v, R)$ be a directed cycle in an n-rotator graph and g_k be the highest dimension generator in R. A node $v' = s_1s_2s_3\ldots s_n$ in the directed cycle C is a *Cut*-node in the directed cycle C if $s_1 = \min \{s_1, s_2, \ldots, s_k\}$.

Lemma 5. Let $C = \pi(v, R)$ be an arbitrary directed cycle in an n-rotator graph, where $v = s_1s_2s_3\ldots s_n$ and g_k is the highest dimension generator in R. There exists at least one *Cut*-node in the directed cycle C.

Proof. Let $s_i = \min \{s_1, s_2, \ldots, s_k\}$. By Lemma 3 and Lemma 4, we know that there exists at least one node $v' = s'_1s'_2s'_3\ldots s'_n$ in the directed cycle C, where $s'_1 = s_i$, and all symbol $s'_j \in \{s_1, s_2, \ldots, s_k\}$ for $1 \le j \le k$. Because $s'_1 = s_i = \min \{s_1, s_2, \ldots, s_k\} = \min \{s'_1, s'_2, \ldots, s'_k\}$, v' is a *Cut*-node in the directed cycle C. □

3 Algorithm

This section proposes two efficient algorithms for finding an FAS in a rotator graph. The first algorithm for efficiently finding an FAS in rotator graphs whose edge g_2 is represented by a directed edge is shown as follows:

Algorithm *FAS-Rotator*-I
Let FAS_n be an FAS in an n-rotator graph
For each node $v = s_1s_2s_3\ldots s_n$ in an n-rotator graph do

```
    flag = true
    For i = 2 to n
        If s1 < si and flag
            Add (v, gi) to FASn
        Else
            flag = false
        Endif
    Enddo
Enddo
```

Because there are n! nodes in an n-rotator graph, Algorithm *FAS-Rotator*-I mainly contains a loop executed n! times and checks all outgoing arcs for each node in this loop. If any outgoing arc satisfies the rule, this algorithm adds the arc to the feedback arc set. In this algorithm, we assume that the single node inside a loop is v, and we initialize a variable *flag* to be true first. Then this algorithm uses a loop to sequentially check from the 2^{nd} to the last symbol of v to see whether it is smaller than the first symbol of v. If s_k, $2 \le k \le n$, is the leftmost symbol that is smaller than s_1, this algorithm sets *flag* to false in order to indicate that s_1 is the smallest symbol among the leftmost $(k – 1)$ symbols of v. Thus arcs (v, g_2), (v, g_3),…, (v, g_{k-1}) are added to the feedback arc set. For example, let $v = \underline{243}15$; then $(24315, g_2)$ and $(24315, g_3)$ are added to FAS_5.

Theorem 1. Algorithm *FAS-Rotator*-I can find an FAS in an n-rotator graph correctly, where edge g_2 is represented as a directed edge.

Proof. Let $C = \pi(v, R)$ be a directed cycle in an n-rotator graph and g_k be the highest dimension generator in R. By Lemma 1 and Lemma 5, there exists a directed cycle $C_1 = \pi(v_1, R_1) = C$, where $v_1 = s_1s_2s_3\ldots s_n$ is a *Cut*-node in the directed cycle C and $s_1 = \min \{s_1, s_2, \ldots, s_k\}$. Let the first generator of R_1 be g_j. Because g_k is the highest dimension generator among R_1, there only exist g_j for $2 \le j \le k$. Let all (v_1, g_j)'s, for $2 \le j \le k$, be denoted by *F*-arcs. Because $s_1 = \min \{s_1, s_2, \ldots, s_k\}$, Algorithm *FAS-Rotator*-I adds the *F*-arcs to FAS_n and removes the *F*-arcs from the n-rotator graph. Thus, directed cycle C_1 must be disconnected at the *Cut*-node v_1. Since Algorithm *FAS-Rotator*-I adds all *F*-arcs's to FAS_n and removes the FAS_n from the n-rotator graph, all directed cycles in the n-rotator graph become disconnected. □

For instance, let directed cycle $C_1 = \pi(\underline{423}1, \boldsymbol{g_3}\, g_2\, \boldsymbol{g_3}\, g_2) = 4231 \xrightarrow{(3)} \underline{\mathbf{234}}1 \xrightarrow{(2)} 3241 \xrightarrow{(3)} \underline{\mathbf{243}}1$

$_{(2)}\rightarrow$ 4231 and directed cycle $C_2 = \pi$(4231, g_2 $\mathbf{g_3}$ g_2 $\mathbf{g_3}$) = 4231 $_{(2)}\rightarrow$ **2431** $_{(3)}\rightarrow$4321 $_{(2)}\rightarrow$3421 $_{(3)}\rightarrow$ 4231. Because g_3 is the highest dimension generator sequence, symbols 4, 2, and 3 must be shifted to the first position in at least one of the nodes in the directed cycle *C*. Since symbol 2 is the smallest symbol among 4, 2 and 3, Algorithm *FAS-Rotator*-I adds (2341, g_2), (2341, g_3), (2431, g_2) and (2431, g_3) to FAS_4, where these arcs are removed from the 4-rotator graph. Obviously, the directed cycle C_1 is disconnected at *Cut*-node's **23**41 and **24**31, and the directed cycle C_2 is disconnected at *Cut*-node **24**31.

Theorem 2. The size of the feedback arc set for an *n*-rotator graph generated by Algorithm *FAS-Rotator*-I is $\sum_{2\le i\le n} n!/i$.

Proof. We will prove this theorem by induction on the size *n* of the rotator graph.

Case *n*=3. Algorithm *FAS-Rotator*-I generates FAS_3 = {(123, g_2), (123, g_3), (132, g_2), (132, g_3), (231, g_2)}. Hence $|FAS_3|$ = 3!/2 + 3!/3 = 5.

Case *n*=4. Algorithm *FAS-Rotator*-I generates FAS_4 = {(1234, g_2), (1234, g_3), (1234, g_4), (1243, g_2), (1243, g_3), (1243, g_4), (1324, g_2), (1324, g_3), (1324, g_4), (1342, g_2), (1342, g_3), (1342, g_4), (1423, g_2), (1423, g_3), (1423, g_4), (1432, g_2), (1432, g_3), (1432, g_4), (2314, g_2), (2341, g_2), (2341, g_3), (2413, g_2), (2431, g_2), (2431, g_3), (3412, g_2), (3421, g_2)}. Hence $|FAS_4|$ = 4!/2 + 4!/3 + 4!/4 = 26.

Suppose case *n*=*k* that $|FAS_k|$ = *k*!/2 + *k*!/3 + … + *k*!/*k* is true.

Now we would like to prove case *n*=*k*+1 that $|FAS_{(k+1)}| = \sum_{2\le i\le (k+1)} (k+1)!/i$. A (*k*+1)-rotator graph is constructed by (*k*+1) *k*-rotator graphs, so the total number of g_j's in $FAS_{(k+1)}$ for $2 \le j \le k$, is (*k*+1)(*k*!/2 + *k*!/3 + … + *k*!/*k*). Let $v = s_1s_2s_3\ldots s_{k+1}$ be a node in a (*k*+1)-rotator graph. Because symbol 1 is the smallest symbol of $s_1, s_2,\ldots$, and s_{k+1}, Algorithm *FAS-Rotator*-I does not add (v, g_{k+1}) to $FAS_{(k+1)}$, where s_j = 1 for $2 \le j \le k$. Consequently, arcs g_{k+1}'s in $FAS_{(k+1)}$ are (v, g_{k+1})'s where $v \in V_{1,1}$, and the total number of arcs g_{k+1}'s is *k*! Therefore,

$|FAS_{(k+1)}| = \underline{(k+1)(k!/2 + k!/3 + \ldots + k!/k)} + \underline{k!}$
$= (k+1)!/2 + (k+1)!/3 + \ldots + (k+1)!/k + k!$
$= (k+1)!/2 + (k+1)!/3 + \ldots + (k+1)!/k + (k+1)!/(k+1)$
□

Theorem 3. The size of the feedback arc set for a rotator generated by Algorithm *FAS-Rotator*-I is minimum.

Proof: Let *v* be a node in an *n*-rotator graph and δ be a generator sequence that is *i* successive generators g_i's for $2 \le i \le n$. Since $v = v * \delta$, *i* successive arcs g_i's construct a directed cycle, denoted by G_i-cycle, for $2 \le i \le n$. Any arc g_i in an *n*-rotator graph only belongs to one of the G_i-cycle's and *i* successive arcs g_i's construct a G_i-cycle, so there are $n!/i$ G_i-cycle's, and each of these G_i-cycle's is arc disjoint, for $2 \le i \le n$. Then, an *n*-rotator graph contains *n*!/2, *n*!/3, *n*!/4, …, and *n*!/*n* arc disjoint directed cycles of length 2, 3, 4, …, and *n*, respectively. In order to break these directed cycles, we have to remove at least one arc from each disjoint directed cycle. Thus, we have to remove at least *n*!/2 + *n*!/3 + *n*!/4 + …+ *n*!/*n* = $\sum_{2\le i\le n} n!/i$ arcs. The result of Algorithm *FAS-Rotator*-I is equivalent to this low bound, and therefore the result is minimum. □

Subsequently, we discuss how to find the FAS in a rotator graph whose edge g_2 is represented by an undirected edge as follows:

Now we consider the case that each edge g_2 is treated as an undirected edge in a rotator graph, and thus g_2 g_2 is not a generator sequence in a directed cycle. Let FAS_3 be a FAS generated by Algorithm *FAS-Rotator*-I. If we remove the FAS_3 except g_2's from the 3-rotator graph as shown in Fig. 2, the directed cycle, 123 $_{(2)}\rightarrow$ 213 $_{(3)}\rightarrow$ 132 $_{(2)}\rightarrow$ 312 $_{(3)}\rightarrow$ 123, exists in the 3-rotator graph. Nevertheless, we remove the FAS_3 from the 3-rotator graph, and no directed cycle exists in the 3-rotator graph because arcs (123, g_2) and (132, g_2) have been removed. In order to generate a minimum FAS, only one of the arcs (123, g_2) and (132, g_2) has to be removed from the 3-rotator graph, and we choose the (123, g_2).

The modified algorithm for finding FAS in rotator graphs is as follows:

Algorithm *FAS-Rotator*-II

```
Let FAS'n be an FAS in an n-rotator graph
For each node v = s1s2s3…sn in an n-rotator graph
do If s1 < s2
     flag = true
     Phase 1:
        For i = 3 to n
           If s1 < si and flag
                 Add (v, gi) to FAS'n
           Else
                 flag = false
           Endif
        Enddo
     Phase 2:
   Endif If s1 < s2 < s3 Add (v, g2) to FAS'n
Enddo
```

Definition 8. RG_n denotes an *n*-rotator graph; FAS_n denotes the FAS generated by Algorithm *FAS-Rotator*-I for the RG_n; *FAS*-1'$_n$ denotes the arc set generated by phase1 of Algorithm *FAS-Rotator*-II for the RG_n; *FAS*'$_n$ denotes the FAS generated by Algorithm *FAS-Rotator*-II for the RG_n.

Theorem 4. There only exists the type of directed cycle $C = \pi(v, R)$ in the $(RG_n$ - $FAS\text{-}1'_n)$, where g_k is the highest dimension generator among $R = \boldsymbol{g_2}\ g_a\ \boldsymbol{g_2}\ g_b\ldots$ for $3 \leq a, b \leq k$, and $v = s_1s_2s_3\ldots s_n$ is a *Cut*-node in the directed cycle C.

Proof. By Theorem 1, the directed cycle C must be disconnected at the *Cut*-node v in the $(RG_n$ – FAS) because all (v, g_j)'s in the $(RG_n$ – FAS) with $2 \leq j \leq k$, are removed. Let $G2_n$ be an arc set of all arcs g_2's in the RG_n. Similarly, the directed cycle C must be disconnected at the *Cut*-node v in the $(RG_n – FAS\text{-}1'_n – G2_n)$. If we do not remove any arc g_2 from the RG_n, then there must exist an arc g_2 at the *Cut*-node v in the $(RG_n$ - $FAS\text{-}1'_n)$, and the directed cycle C is not disconnected at the *Cut*-node v in the $(RG_n – FAS\text{-}1'_n)$ by (v, g_2) existing. Let $C' = \pi(v', R')$ be a directed cycle in the $(RG_n$ - $FAS\text{-}1'_n)$, where g_k be the highest dimension generator in R'. By Lemma 1 and Lemma 5, there exists a directed cycle $C = \pi(v, R) = C'$. Thus, only $\pi(v, g_2\ldots)$'s exist in the $(RG_n – FAS\text{-}1'_n)$. Since v is a *Cut*-node in the directed cycle C, s_1 is the smallest symbol among $s_1, s_2, \ldots$ and s_k. Let $v_1 = v * g_2$, and then s_1 is the second symbol of v_1. Because s_1 is the smallest symbol among $s_1, s_2, \ldots$ and s_k, the first symbol of v_1 is greater than s_1 and then $\pi(v_1, g_a\ldots)$'s exist in the $(RG_n – FAS\text{-}1'_n)$, where $3 \leq a \leq k$. Let $v_2 = v_1 * g_a$, and then s_1 is the first symbol of v_2. Because s_1 is the first symbol of v_2 and s_1 is the smallest symbol among $s_1, s_2, \ldots$ and s_k, v_2 is also a *Cut*-node in the directed cycle C, and only $\pi(v_2, g_2\ldots)$'s exist in the $(RG_n – FAS\text{-}1'_n)$. Let $v_3 = v_2 * g_2$, and then s_1 is the second symbol of v_3. Because s_1 is the smallest symbol among $s_1, s_2, \ldots$ and s_k, the first symbol of v_3 is greater than s_1 and then $\pi(v_3, g_b\ldots)$'s exist in the $(RG_n – FAS\text{-}1'_n)$, where $3 \leq b \leq k$. By Lemma 1, $\pi(v, \boldsymbol{g_2}\ g_a\ \boldsymbol{g_2}\ g_b\ldots) = \pi(v_1, g_a\ldots) = \pi(v_2, g_2\ldots) = \pi(v_3, g_b\ldots) = \ldots$. Thus, only $\pi(v, \boldsymbol{g_2}\ g_a\ \boldsymbol{g_2}\ g_b\ldots)$'s exist in the $(RG_n – FAS\text{-}1'_n)$, where v is a *Cut*-node in the directed cycle C. □

Definition 9. A node $v = s_1s_2s_3\ldots s_n$ in an n-rotator graph is a G-permutation if $s_1 < s_2 < s_3$, where $n > 2$.

Theorem 5. Let $C = \pi(v, R)$ be a directed cycle in an n-rotator graph, where g_k is the highest dimension generator among $R = \boldsymbol{g_2}\ g_a\ \boldsymbol{g_2}\ g_b\ldots$ for $3 \leq a, b \leq k$, and $v = s_1s_2s_3\ldots s_n$ is a *Cut*-node in the directed cycle C. There exists at least one node which is a *Cut*-node in the directed cycle C and a G-permutation.

Proof. By Lemma 4, all symbols $s_1, s_2,\ldots,$and s_k must be shifted to the first symbol position in at least one of the nodes in the directed cycle C. Let $v_1 = v * \boldsymbol{g_2} = s_2s_1s_3\ldots s_n$, $v_2 = v_1 * g_a = s_1s_3\ldots s_n$, $v_3 = v_2 * \boldsymbol{g_2} = s_3s_1\ldots s_n$, and … . In other words, all symbols $s_2, \ldots,$ and s_k must be shifted to the second symbol position in at least one of the nodes in the directed cycle C and then are shifted to the first symbol position in one of the nodes in the directed cycle C by generator g_2. In addition, the symbol s_1 must be shifted to the first position in the node when the symbol s_j is shifted to the second symbol position in one the nodes in the directed cycle C for $2 \leq j \leq k$. Let s_i be the smallest symbol of $s_2, s_3,\ldots,$ and s_k. Since v is a *Cut*-node in the directed cycle C, s_1 is the smallest symbol among $s_1, s_2, \ldots$ and s_k. Thus, we know that at least there exists a node $v' = s_1s_i\ldots$ in the directed cycle C, where $s_1 = \min\{s_1, s_2, s_3,\ldots$ and $s_k\}$, $s_i = \min\{s_2, s_3,\ldots$ and $s_k\}$, the j^{th} symbol of $v' \in \{s_1, s_2,\ldots,s_k\}$ for $1 \leq j \leq k$, and $k > 2$. □

Theorem 6. Algorithm *FAS-Rotator*-II can find an FAS in an n-rotator graph correctly, where edge g_2 is presented as an undirected edge.

Proof. By Theorem 4, we know that there only exists the type of directed cycle $C = \pi(v, R)$ in the $(RG_n$ - $FAS\text{-}1'_n)$, where g_k be the highest dimension generator among R, and $R = \boldsymbol{g_2}\ g_a\ \boldsymbol{g_2}\ g_b\ldots$ for $3 \leq a, b \leq k$, and v is a *Cut*-node in the directed cycle C. Let *Cut*-*G*-node be a *Cut*-node in the directed cycle C and a G-permutation. By Theorem 5, we know that there exists at least one *Cut*-*G*-node in the directed cycle C. Because there exist (v, g_2)'s in the *Cut*-node's, the directed cycle C is not disconnected at the *Cut*-node's in the $(RG_n – FAS\text{-}1'_n)$. Let $G2$-arcs be an arc set of arcs g_2's of all G-permutation's in the RG_n. Thus, there exists at least one *Cut*-*G*-node whose arc g_2 is removed from the directed cycle C in the $(RG_n$ - $FAS\text{-}1'_n$ - $G2$-arcs), and the directed cycle C must be disconnected at the *Cut*-*G*-node in the directed cycle C in the $(RG_n$ - $FAS\text{-}1'_n$ - $G2$-arcs). Consequently, Algorithm *FAS-Rotator*-II adds the $FAS\text{-}1'_n$ and the $G2$-arcs to FAS'_n and removes the FAS'_n from the n-rotator graph, and then all directed cycles in the $(RG_n$ - $FAS\text{-}1'_n$ - $G2$-arcs) become disconnected. □

For example, $\pi(\underline{245}31, g_2\ \boldsymbol{g_4}\ g_2\ g_4\ g_2\ g_4)$ is a directed cycle of **245**31 $_{(2)}\!\rightarrow$ 42531 $_{(4)}\!\rightarrow$ 25341 $_{(2)}\!\rightarrow$ 52341 $_{(4)}\!\rightarrow$ **234**51 $_{(2)}\!\rightarrow$ 32451 $_{(4)}\!\rightarrow$ 24531, and all arcs g_i's of nodes **245**31 and **234**51 are in FAS'_5 and removed for $2 \leq i \leq 4$. Thus, the directed cycle in this example is disconnected at these above nodes.

Theorem 7. The size of the feedback arc set for an n-rotator graph generated by Algorithm *FAS-Rotator*-II is $(n!/6 + \sum_{3\leq i\leq n} n!/i)$, which is minimum.

Proof. Let $v = s_1s_2s_3\ldots s_n$ be a node in an n-rotator graph. $|FAS_n| = (\underline{n!/2} + \sum_{3\leq i\leq n} n!/i)$, where the g_2's of FAS_n is the g_2's of v for $s_1 < s_2$. Because $(|\{v \mid \text{for } s_1 < s_2\}| \;/\; |\{v\}|) = 1/2$, $|\{\forall\, g_2 \in FAS_n\}| = n!/2$. The g_2's of FAS'_n is the g_2's of v for $s_1 < s_2 < s_3$. Because $(|\{v \mid \text{for } s_1 < s_2 < s_3\}| \;/\; |\{v\}|) = 1/6$, $|\{\forall\, g_2 \in FAS'_n\}| = n!/6$. Therefore, $|\{FAS'_n\}| = (\underline{n!/6} + \sum_{3 \le i \le n} n!/i)$. As proof in Theorem 3, there are $n!/i$ arc disjoint directed cycles of arc g_i in the n-rotator graph for $3 \le i \le n$. Then, the n-rotator graph contains $n!/3$, $n!/4$, …, and $n!/n$ arc disjoint directed cycles of length 3, 4, …, and n, respectively. To break these directed cycles, we have to remove at least one arc from each disjoint directed cycle. Thus, we have to remove at least $n!/3 + n!/4 + \ldots + n!/n = \sum_{3 \le i \le n} n!/i$ arcs. Subsequently, we discuss the least number of arcs g_2's which have to be removed from the n-rotator graph. The smallest directed cycle discussed in theorem 4 is the directed cycle $\pi(v, g_2\, g_3\, g_2\, g_3)$. There are three directed cycles $\pi(v, g_2\, g_3\, g_2\, g_3)$'s in a 3-rotator graph, and they are $C_1 = 123 \xrightarrow{(2)} 213 \xrightarrow{(3)} 132 \xrightarrow{(2)} 312 \xrightarrow{(3)} 123$, $C_2 = \underline{\mathbf{123}} \xrightarrow{\mathbf{(3)}} 231 \xrightarrow{(2)} 321 \xrightarrow{(3)} 213 \xrightarrow{(2)} 123$ and $C_3 = \underline{\mathbf{132}} \xrightarrow{\mathbf{(3)}} 321 \xrightarrow{(2)} 231 \xrightarrow{(3)} 312 \xrightarrow{(2)} 132$, respectively. All arcs g_3 in the three directed cycles C_1, C_2 and C_3 are disjoint. Furthermore, there are two disjoint directed cycles of arc g_3 also in the 3-rotator graph, and they are $C'_1 = \underline{\mathbf{123}} \xrightarrow{\mathbf{(3)}} 231 \xrightarrow{(3)} 312 \xrightarrow{(3)} 123$ and $C'_2 = 213 \xrightarrow{(3)} \underline{\mathbf{132}} \xrightarrow{\mathbf{(3)}} 321 \xrightarrow{(3)} 213$, respectively. We have to remove one arc g_3 from each disjoint directed cycle of arc g_3 in the 3-rotator graph. Since two arcs g_3's have to be removed from the 3-rotator graph and all arcs g_3 in the three directed cycles C_1, C_2 and C_3 are disjoint, at most two of the three directed cycles C_1, C_2 and C_3 are disconnected by these two arcs g_3's removed. In other words, at least one of the three directed cycles C_1, C_2 and C_3 is still connected. For example, the two arcs $(123, g_3)$ and $(132, g_3)$ are removed from the directed cycles C'_1 and C'_2, respectively. Because these two arcs $(123, g_3)$ and $(132, g_3)$ are also in the directed cycles C_2 and C_3 respectively, no arc g_2 has to be removed from these two directed cycles. Thus, only one arc g_2 has to be removed from the directed cycle C_1, and at least one arc g_2 has to be removed from a 3-rotator graph. There are $n!/6$ disjoint 3-rotator graphs in an n-rotator graph, and each disjoint 3-rotator graph is symmetric, so at least $n!/6$ arcs g_2's have to be removed from the n-rotator graph. Consequently, the minimum size of the FAS for the n-rotator graph is $(\underline{n!/6} + \sum_{3 \le i \le n} n!/i)$, which is equivalent to our result. □

Theorem 8. For an n-rotator graph with $n!$ nodes, Algorithm *FAS-Rotator*-I / Algorithm *FAS-Rotator*-II can be done in $O(nn!)$ time.

Proof: Since there are $n!$ nodes in an n-rotator graph and Algorithm *FAS-Rotator*-I / Algorithm *FAS-Rotator*-II checks $(n - 1)$ arcs at most for each node, thus the time complexity is $O(nn!)$. □

4 Conclusions

This paper presents two efficient algorithms for finding a minimum FVS in an n-rotator graph in $O(nn!)$ time. In other words, these algorithms derive the optimal result with linear time complexity in terms of the number of arcs in the rotator graph.

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